



Indian Institute of Technology Kanpur
Department of Mathematics and Statistics
Introduction to Bayesian Analysis (MTH422A)
Assignment 1, Due date: January 31, 2024, Wednesday

Answers should be provided neatly. At least some of the questions cannot be answered without necessary rough work. All such rough work should be attached with the original submission, at the back of the original submission but not written together. If rough work is not shown and you cannot justify that the final answer can be thought very straightforwardly, you will be penalized for cheating. In case of cheating, all students involved will get zero. For R codes, please submit the Markdown file (.Rmd) as well as its PDF output. Thus, there should be a total of four files to be submitted—one PDF with neat solutions, one PDF with rough work, one .Rmd file, and one .pdf which is the output of the .Rmd file.

1. Suppose $X_n; n \geq 1$ is a sequence of IID random variables satisfying $X_i \stackrel{IID}{\sim} \text{Bernoulli}(\theta), \theta \in (0, 1)$. Derive the distribution of $Y_N = \sum_{n=1}^N X_n$. Also, $\bar{Y}_N = N^{-1}Y_N$. What is the support of $\bar{Y}_N$? Numerically compare the CDF of $\bar{Y}_N$ and its approximate distribution based on CLT (in terms of Kolmogorov-Smirnov distance) using R for $N = 10, 50, 500$. (0.5+1.5 = 2 points)
2. Suppose $X_n; n \geq 1$ is a sequence of IID random variables satisfying $X_i \stackrel{IID}{\sim} \text{Gamma}(\alpha, \lambda)$. Let $\bar{X}_N = N^{-1} \sum_{n=1}^N X_n$. Find the distribution of $\bar{X}_N$. (1 point)
3. Suppose X is a random variable with density function $f_X(x) = 2K_0(2\sqrt{x}), x \in (0, \infty)$, where K_0 is a Bessel function (see the `besselK(x, nu = 0)` in R). Here

$$K_0(2x) = \frac{1}{2} \int_0^\infty y^{-1} \exp[-y - x^2/y] dy.$$

Using analytical as well as numerical integration (using R), check that f_X is a valid PDF. Write down a hierarchical representation of X , i.e., $X|Y \sim F, Y \sim G$ where F and G are standard well-known distributions. (0.5+0.5+1 = 2 points)

4. Suppose $X_1 \sim \text{Normal}(0, 1)$ and $X_t|X_{t-1} \sim \text{Normal}(\rho X_{t-1}, 1 - \rho^2), t = 2, \dots, T$. Find the joint distribution of $\mathbf{X} = (X_1, \dots, X_T)'$. Write down the conditional distribution of X_t given $X_1, \dots, X_{t-1}, X_{t+1}, \dots, X_T$ for some $t \in \{2, 3, \dots, T-1\}$. (1+1 = 2 points)
5. Find a hierarchical representation of the vector $\mathbf{X}$ so that the joint distribution of $\mathbf{X}$ is multivariate normal with zero mean, unit marginal variances, and all off-diagonal elements equal to $\rho \in (0, 1)$. The hierarchical structure should be based on all univariate components. (1 point)
6. Suppose $Z_t|Y_t \stackrel{Indep}{\sim} \text{Normal}(Y_t, 1), t = 1, \dots, T$, and $Y_1, \dots, Y_T|X \stackrel{IID}{\sim} \text{Poisson}(X)$, and $X \sim \text{Gamma}(a, b)$. Using Bayes theorem, obtain the conditional distribution of X given $Y_1, \dots, Y_T$. (1 point)
7. Using R, draw 10^4 IID samples from the uniform distribution within the curvature $(x^2 + y^2 - 1)^3 \leq x^2 y^3$ using acceptance-rejection sampling as efficiently as possible and plot them. Using the R package `tictoc`, check the time required to draw these samples. (1 point)